Shunri Zheng

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Professional Summary

PhD researcher in Machine Learning & Operations Research specializing in neural-network generalization theory (PAC–Bayes), predictive reconstruction, and autoregressive sequence modeling. Build GPU-efficient, reproducible training/evaluation pipelines in PyTorch + Hugging Face with bound-based metrics and large-scale experiment tracking (W&B).

Technical Skills

LLM/Deep Learning: PyTorch, Hugging Face (Transformers/Datasets/TRL/Accelerate), Weights & Biases, PPO/RLHF
Data/Tools: Python, SQL, R, Julia, Git

Education

University of Illinois Urbana-Champaign — Aug 2025 - July 2030 (Expected)
Ph.D. in Industrial Engineering — *Advisor: Prof. Yunzong Xu*
Columbia University in the City of New York — M.A. in Statistics
The Chinese University of Hong Kong, Shenzhen — B.Sc. in Statistics (Data Science Stream)

Professional Experience

Columbia Business School — New York, NY
Research Assistant - Machine Learning and Operations Research — Dec 2023 - Dec 2024

- Built reproducible training/evaluation pipelines for autoregressive transformer models applied to queueing dynamics and policy optimization; modularized data ingestion, simulation, and metrics to support scenario stress-tests and reruns.
- Ran and tracked large experiment suites with Weights & Biases, automated sweeps, and dataset versioning to enable fast iteration and reproducible baselines across collaborators.
- Processed and analyzed de-identified healthcare and call-center operational event logs; built reusable reporting notebooks and dashboards for model debugging and error analysis.

Research Projects

Predictive Reconstruction and Generalization for Neural Networks — Sept 2025 - Present
Advisor: Prof. Yunzong Xu — University of Illinois Urbana-Champaign

- Developed a PAC–Bayes framework for neural-network generalization through predictive reconstruction and layer refitting; derived guarantees independent of original-network size under explicit structural assumptions.
- Evaluated importance-sampling certificates across 42 ReLU networks on digit and two-moons tasks; obtained nonvacuous classification-error upper bounds for all 30 clean-label models, with tighter bounds than tested compression-code and no-refit baselines on 28/30 and 30/30 models, respectively.
- Designed controlled clean- versus random-label experiments showing that low-rank features alone do not ensure successful predictive reconstruction on unseen inputs.

Data-Driven Stochastic Modeling Using Autoregressive Sequence Models — Feb 2024 - Sept 2025
Advisor: Prof. Hongseok Namkoong, Prof. Jing Dong — Columbia University

- Modeled operational event sequences with autoregressive transformers to learn queueing dynamics and optimize staffing policies; validated against oracle/posterior baselines across thousands of simulated scenarios.
- Built end-to-end transformer training pipeline in PyTorch for M/M/5 queues with heterogeneous arrival/service rates; standardized configs, seeds, and evaluation for reproducible experiments.
- Performed counterfactual simulations with time-varying demand, achieving 90% accuracy in resource allocation on unseen environments; analyzed sensitivity to stochastic shocks for robust policy selection.

Triply Robust Causal Estimation For Continuous Treatments — Jun 2024 - Present
Advisor: Prof. Hongseok Namkoong, Prof. Yunbei Xu — Columbia University

- Developed a triply robust estimator for continuous treatment effects with bandwidth selection, improving RMSE by 50% over prior methods.
- Proved asymptotic normality to enable valid inference and policy evaluation for pricing interventions.
- Implemented large-scale experiments on e-commerce transaction data, achieving 30% lower RMSE and integrating diagnostics for overlap and sensitivity analysis.

Publications & Manuscripts

- "Data-Driven Stochastic Modeling Using Autoregressive Sequence Models: Translating Event Tables to Queueing Dynamics," *Shunri Zheng, Daksh Mittal, Hongseok Namkoong, Jing Dong* (Submitted to **Management Science**, 2025). [arXiv]
- "Triply Robust Causal Estimation for Continuous Treatments," *Yunbei Xu, Shunri Zheng, Hongseok Namkoong* (In preparation for **Operations Research**).

Honors and Awards

- The 2023 Chair's List Award in Academic Achievement, Columbia University
- Dean's List (2018-20, 2021-22), The Chinese University of Hong Kong, Shenzhen
- Academic Performance Scholarship (2019-20), The Chinese University of Hong Kong, Shenzhen